

K4 EEId Solution and Procedure

From Ibn Şifre to Yahoo Kryptos Group:

Bismillah, Let us begin:

Part 1

Applying Mask Step:

Start with K3 and arrange it in 7 rows by 48 columns as shown in Fig 1-1.

	1	2	3	4
	123456789012345678901234567890123456789012345678			
1	ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIACHTNREYULDSL	LSLL		
2	NOHSNOSMRWXMNETPRNGATIHNRARPESLNNELEBLPIIACA	EWMT		
3	WNDITEENRAHCTENEUDRETNHAEOETFOLSEDTIWENHAEIOY	TEY		
4	QHEENCTAYCREIFTBRSPAMHHEWENATAMATEGYEERLBTEE	FOAS		
5	FIOTUETUAEOTOARMAEERTNRTIBSEDDNIAAHTTMSTEWPIE	ROA		
6	GRIEWFEBAECTDDHILCEIHSITEGOEAOSSDDRYDLORITRKL	MLEH		
7	AGTDHARDPNEOHMGFMFEUHEECDM RIPFEIMEHNLSSTTRTV	DOHW		

Fig. 1-1, K3

Flip K3 about the columns so that column 48 is first and column 1 is last, as shown in Fig 1-2. This is done only to make the technique easier to visualize. Call it K3'.

	4	3	2	1
	876543210987654321098765432109876543210987654321			
1	LLSLLSDLUYERNTHCAIANHSYDIBIOETPCOEHRSLNHORHAYDNE			
2	TMWEACAIIPLBELENNLSEPRARNHITAGNRPTENMXWRMSONSHON			
3	YETYOIEAHNEWITDESLOFTEOEAHNTERDUENETCHARNEETIDNW			
4	SAOFEETBLREEYGETAMATANEWEEHMAPSRBTFIERCYATCNEEHQ			
5	AOREIPWETSMTTHAAINDDESBITRNTREEAMRAOTOEAEUTEUTOIF			
6	HEMLKRTIROLDYRDDSOAEOGETISHIECLIHDDTCEABEFWEIRG			
7	WHODVTRTTSSLNHEMIEFFIRMDCEEHUEFMFGMHOENPDRAHDTGA			

Fig. 1-2, K3'

Using the END - YAR (RAY) clue, place K3' in the form of a transverse EM wave with a period of 16 characters and an amplitude of 11 characters, as shown in Fig 1-3. Call it W-K3'

	4	3	2	1		
	876543210987654321098765432109876543210987654321					
4	LLSD	RNTH	NHSY	OETP	RSLN	AYDN
5	EACA	BELE	EPRA	TAGN	NMXW	NSHO
6	YOIE	WITD	FTEO	TERD	TCHA	TIDN
7	FEET	EYGE	TANE	MAPS	IERC	NEEH
1	LLSEIPWLUYETTHACAIADIESBDIBITREECOEHOTOEHORHUTOIE					
2	TMWMLKRIIPLLDYRNNLSEOGRNHIEHIECRPTEDTCERMSOWEIRN					
3	YETDVTRAHNELNHEESLOPIRMEAHNHUEFUENEHOENRNEEHDTGW					
4	SAO	BLRE	TAMA	WEHH	RBTF	YATC Q
5	AOR	ETSM	AIND	ITRN	AMRA	AUTE F
6	HEL	TIRO	DDSO	ETIS	LIHD	ABEF G
7	WHO	TTSS	MIEF	DCEE	MFGM	PDRA A

Fig. 1-3, W-K3'

As an aside, notice in Row 1, Columns 41 to 48 of W-K3', as shown in Fig. 1-4, spell out 'L WPIES LL'. Maybe it means 'L SPIES LL'; spies surrounded by L's.

	4		4
	876543210		012345678
1	LLSEIPWL	Or	1 LWPIESLL

Fig. 1-4

For brevity sake, display only rows 25 through 48 of EM-K3' as shown in Fig 1-5. Call it W1, and notice the re-ordering of the columns.

	1										2														
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	
4																									
5																									
6																									
7																									
1	L	L	S	D							R	N	T	H							N	H	S	Y	
2	E	A	C	A							B	E	L	E							E	P	R	A	
3	Y	O	I	E							W	I	T	D							F	T	E	O	
4	F	E	E	T							E	Y	G	E							T	A	N	E	
1	L	L	S	E	I	P	W	L	U	Y	E	T	T	H	A	C	A	I	A	D	E	S	B	D	
2	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R	
3	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E	
4	S	A	O								B	L	R	E							T	A	M	A	
5	A	O	R								E	T	S	M							A	I	N	D	
6	H	E	L								T	I	R	O							D	D	S	O	
7	W	H	O								T	T	S								M	I	E	F	

Fig. 1-5, W1

Now rearrange W1 so that the wave is in the opposite phase as shown in Fig. 1-6, W2.

	1										2														
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	
1	L	L	S								L	U	Y	E							C	A	I	A	
2	T	M	W								I	I	P	L							N	N	L	S	
3	Y	E	T								A	H	N	E							E	S	L	O	
4	S	A	O								B	L	R	E							T	A	M	A	
5	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E						A	I	N	D	E
6	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E	
7	W	H	O	F	E	E	T	T	S	S	E	Y	G	E						M	I	E	F	T	
1											E	I	P	W						T	T	H	A		
2											M	L	K	R						L	D	Y	R		
3											D	V	T	R						L	N	H	E		
4											L	L	S	D						R	N	T	H		

Fig. 1-6, W2

OTP, One Time Pad, Decryption Step:

The next step involves treating W1 as a pad and K4 as ciphertext as part of a OTP, one time pad, encryption scheme. In order to recover the plaintext one needs to follow the following decryption algorithm. Assign to each letter of the Kryptos alphabet a number as shown in Fig. 1-7:

K	R	Y	P	T	O	S	A	B	C	D	E	F
0	1	2	3	4	5	6	7	8	9	10	11	12
G	H	I	J	L	M	N	Q	U	V	W	X	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Fig. 1-7, Letter to Number Mapping for OTP

Then, for a letter of the ciphertext and a letter of the pad, subtract the numbers corresponding to the letters. If the resulting number is negative then add 26 to it. Take the resulting number and find its corresponding letter. That letter is the letter of the plaintext.

Examples:

```
ciphertext => NY....
pad         => EL....
-----
plaintext  => BE....
```

Example 1:

```
step 1:
N => 19
E => 11
```

```
step 2:
19 - 11 = 8
```

```
8 => B
plaintext => B
```

Example 2:

```
step 1:
Y => 2
L => 17
```

```
step 2:
2 - 17 = -15
```

```
step 3:
-15 + 26 = 11
```

```
11 => E
plaintext => E
```

For ease, I have provided a python script that automates this in my folder in the Yahoo Kryptos Group Files under personal folders.

Continuing, Arrange K4 in 6 rows so that they look like small "pyramids" on their sides as shown in K4, Fig 1-8.

```

1      OBKRUOXOGHULBSO
2  LIFBBWFLRVQQPRNGKSSO
3      TWTQSJQSSEKZZW

4  ATJKLUDIAWINFB
5  NYPVTTMZFPKWGDKZXTJC
6  DIGKUHUAUEKCAR

```

Fig. 1-8, K4

Take Row 1 of K4 and Row 1, Columns 9 to 23 of W1, Fig. 1-5 (after the LLSEIPWL clue) and apply the OTP for the resulting plain text:

```

K4 R1  OBKRUOXOGHULBSO
W1 R1  UYETTHACAIADESB
-----
      DSIWLLLVSHAWKW

```

And Take Row 2 of K4 and Row 2, Columns 4 to 23 of W1 and apply the OTP for the resulting plain text:

```

K4 R2  LIFBBWFLRVQQPRNGKSSO
W1 R2  MLKRIIPLLDYRNNLSAEOG
-----
      ZXFANBCKDFMNDYANURM

```

And Take Row 3 of K4 and Row 3, Columns 10 to 23 of W1 and apply the OTP for the resulting plain text:

```

K4 R3  TWTQSJQSSEKZZW
W1 R3  NELNHEESLOPIRM
-----
      EFGRMOCKISWDXO

```

And Take Row 4 of K4 and Row 5, Columns 2 to 15 of W2 and apply the OTP for the resulting plain text:

```

K4 R4  ATJKLUDIAWINFB
W1 R5  OREACAETSMBELE
-----
      YPONBHZEROABUW

```

And Take Row 5 of K4 and Row 6, Columns 2 to 21 of W2 and apply the OTP for the resulting plain text:

K4	R5	NYPVTTMZFPKWGDZXTJC
W1	R6	ELYOIETIROWITDDDSOFT

		BERLINHDEXPBCKJIMZTO

And Take Row 6 of K4 and Row 7, Columns 2 to 15 of W2 and apply the OTP for the resulting plain text:

K4	R6	DIGKUHUAUEKCAR
W1	R7	HOFEETTSSEYGE

		VDRIDDLPIOIAQJ

Now arrange the 'pyramids' base to base and get P1 as shown in Fig. 1-9,

1	DSIWLLLVSHAWKW	YPONBHZEROABUW
2	ZXFANBCKDFMNDYANURM	BERLINHDEXPBCKJIMZTO
3	EFGRMOCKISWDXO	VDRIDDLPIOIAQJ

Fig. 1-9, P1

Misspellings Step:

Guess the misspellings as follows in Fig. 1-10.

Don't worry, there are ways to tell if you have the right ones. The rows labeled with P1M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P1 are the plaintext row from the OTP step. The rows labeled with W2 are the rows from the pad text and show the corresponding pad letter where the misspelling occurred in the plaintext.

		1		2			1		2	
	123456789012345678901234		123456789012345678901234			123456789012345678901234				
P1M 1		C	I	B		A	S	N		
P1 R1		DSIWLLVSHAWKW				YPONBHZEROABUW				
W2-14		UYETHACIADESB				OREACAETSMBELE				
P1M 2	K		Q	I	O		S	E	T	S
P1 R2	ZXFANBCKDFMND				BYANURM	BERLINHDEXPBCKJIMZTO				
W2-25	MLKRIIPLLDYRNNLSAEOG					ELYOIETIROWITDDDSOFT				
P1M 3		D	R		R		D	E	C	
P1-R3		EFGRMOCKISWDXO				VDRIDDLPIOIAQJ				
W2-36		NELNHEESLOPIRM				HOFEETTSSEYGE				

Fig. 1-10

Which gives us PT1, our K4 solution to the 97 charters as shown in Fig. 1-11

```

1      DCIWILLVSBHAWKW YPANBHZEROASUN
2 KXFANBCQDFINDBYANORM BERLINSDEXPECTJIMSTO
3      EDGRROCKISRDXO DDRIDDLEIOICQJ

```

Fig. 1-11, PT1

Examining the misspellings, Fig. 1-11, we can find further messages.

		1		2			3		4
	12345678901234567890		1234567890123456789			1234567890123456789			
P1M 1		DCIWILLVSBHAWKW		YPANBHZEROASUN					
P1M 2	KXFANBCQDFINDBYANORM		BERLINSDEXPECTJIMSTO						
P1M 3	EDGRROCKISRDXO		DDRIDDLEIOICQJ						

Fig. 1-12

Read the misspelled letters across by rows in such a way:

P1M 1 <- right to left

P1M 2 -> left to right

P1M 3 <- right to left

To get the following, PT1-MA, as shown in Fig. 1-13.

NSA BIC K Q IO SETS C ED R RD

Fig. 1-13, PT1-MA

Now read the misspelled letters going up-down, left to right to get the following, PT1-MD, as shown in Fig 1-14. (odd columns: up and even columns: down)

K CDQ IRI BRO DASE SEC NTS

Fig. 1-14, PT1-MD

Next, examine the W2 Letters at the positions of the misspellings, Fig. 1-15.

	1	2	3	4
	12345678901234567890	12345678901234567890	12345678901234567890	12345678901234567890
W2-14	UYETHACAI	IADESB	OREACAETS	SMBELE
W2-25	MLKRII	LLDYRNNLSA	EOG	ELYOIETIROWITDDSOFT
W2-36	NELN	HEESLOP	IRM	HOFEETTSSEYGE

Fig. 1-15

Again, read the misspelled letters across by rows in such a way:

W2-14 <- right to left

W2-25 -> left to right

W2-36 <- right to left

To get the following, PT1-WA, as shown in Fig. 1-16.

EEE ITYM LYE TI DOYTH P HE

Fig. 1-16, PT1-WA

Now read the W2 letters from Fig. 1-15, going up-down, left to right to get the following, PT1-WD, as shown in Fig 1-17.

MY LET HYIPE HETT EIYED O

Fig. 1-17, PT1-WD

Part 2

Bismillahi Al Haqq il Ala, let us continue.

Applying Mask Step:

Start with W1, Fig. 1-5, and re-order its rows the following way:

row 4 to 1
row 1 to 2
row 5 to 3
row 2 to 4
row 6 to 5
row 3 to 6
row 7 to 7

Be careful to follow the row numbers in the figure. For example, part of row 4 is on the first line and the other part is on the eighth line. Call this W1', as shown in Fig. 2-1. W1 row numbers are listed first, followed by W1' row numbers.

			1		2	
W1	W1'	1	2	3	4	5
R	R					
2	4		MLKR		LDYR	AEOG
6	5		YOIE		WITD	FTEO
3	6		DVTR		LNHE	PIRM
7	7		FEET		EYGE	TANE
4	1	SA	OLLSDBLRERNTH	TAMANHSY	W	
1	2	LLSEIPWL	UYETTH	ACAI	ADESBD	
5	3	AOREACAETS	MBELEA	INDEPRAI		
2	4	TMW	I IPL	NNLS	R	
6	5	HEL	TIRO	DDSO	E	
3	6	YET	AHNE	ESLO	E	
7	7	WHO	TTSS	MIEF	D	

Fig. 2-1, W1'

Do the same for W2, Fig.1-6. Call this W2', as shown in Fig. 2-2. W2 row numbers are listed first, followed by W2' row numbers.

		1										2													
W2	W2'	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
R	R																								
4	1			SAO				BLRE							TAMA										W
1	2			LLS				LUYE							CAIA										D
5	3			AOR				ETSM							AIND										I
2	4			TMW				I IPL							NNLS										R
6	5			HEL	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E
3	6			Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R
7	7			W	H	O	F	E	E	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E
4	1			LLSD				RNTH							NHSY										
1	2			EIPW				TTHA							DESB										
5	3			EACA				BELE							EPRA										
2	4			MLKR				LDYR							AEOG										

Fig. 2-2, W2'

OTP, One Time Pad, Decryption Step:

Next, arrange parts K4 like shown in Fig. 2-3:

1	OBKRUOXOGHULBSOLIFBBW
2	FLRVQQPRNGKSSOTWTQSJQ
3	SSEKZZWATJKLUDIAWINFB
4	NYPVTTMZFPKWGD
5	KZXTJCDIGKUHUAUEKCAR

Fig. 2-3, K4'

Take Row 1 of K4' and Row 3, Columns 4 to 23 of W1' and apply the OTP decryption algorithm, as before in part 1, for the resulting plain text:

K4' R1	OBKRUOXOGHULBSOLIFBBW
W1' R3	EACAETSMBELEAINDEPRAI

	QRLQDRMGOPTSRLFATCARB

And Take Row 4 of K4' and Row 2, Columns 2 to 14 of W1' and apply the OTP for the resulting plain text:

K4' R4	NYPVTTMZFPKWGD
W1' R2	LLSEIPWLUYETTH

	YEWEIRUBLRINCV

And Take Row 3 of K4' and Row 1, Columns 3 to 23 of W1' and apply the OTP for the resulting plaintext:

K4' R3	SSEKZZWATJKLUDIAWINFB
W1' R1	OLLSDBLRERNTHTAMANHSY

	RIQQILSSNIAGASBIJVOSS

And Take Row 5 of K4' and Row 7, Columns 2 to 20 of W2' and apply the OTP for the resulting plain text:

K4' R5	KZXTJCDIGKUHUAUEKCAR
W2' R7	WHOFEEETTSSEYGEMIEFT

	PENMOXSECQIPNQDNEXUW

The plain text is backwards, so arrange them as in P2 shown in Fig. 2-4.

1	BRAC	TAFLRSTPOGMRDQLRQ	VCNIRLBURIEWEY
2	SSOVJIBSAGAINSSLIQQIR	WUXENDQNPIQCESXOMNEP	

Fig. 2-4, P2

Misspellings Step:

Guess the misspellings as follows in Fig 2-5. The rows labeled with P2M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P2 are the plaintext row from the OTP step. The rows labeled with W' are the row from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

	2	1		2	1
	432109876543210987654321		432109876543210987654321		
P2M 1		O A L A		S T D	
P2 R1		BRAC	TAFLRSTPOGMRDQLRQ		VCNIRLBURIEWEY
W'-1		IARPEDNIAEIEBMSTEACAE		HTTEYULWPIESLL	
P2M 2		S M T E		C H E O	
P1 R2		SSOVJIBSAGAINSSLIQQIR		WUXENDQNPIQCESXOMNEP	
W'-2		YSHNAMATHTNRERLBDSSLLO		TFEIMEGYESSTTTEEFOWH	

Fig. 2-5

Which gives us PT2, our K4 part 2 solution as shown in Fig 2-6.

BRAC~~T~~OFLAS~~T~~LOGARDQLRQ SCN~~T~~RLBURIEDEY
SSOSJIMSAGAINSTLIEQIR CUXENDQHPIECESXOONEP

Fig. 2-6, PT2

Examining the misspellings, Fig 2-6, we can find further messages.

	2	1		2	1																							
	4	3	2	1	0	9	8	7	6	5	4	3	2	1	4	3	2	1	0	9	8	7	6	5	4	3	2	1
P2M 1	BRAC T OFLAS T LOGARDQLRQ														SCN T RLBURIEDEY													
P2M 2	SSOSJIMSAGAINSTLIEQIR														CUXENDQHPIECESXOONEP													

Fig 2-6

Read the misspelled letters across by rows in such a way:

P2M 1 -> left to right

P2M 2 -> left to right

To get the following, PT2-MA, as shown in Fig. 2-7.

O A LAST DS MT ECHEO

Fig. 2-7, PT2-MA

Now read the misspelled letters going down, left to right to get the following, PT2-MD, as shown in Fig 2-8.

DOETH SCE T ALAMOS

Fig. 2-8, PT2-MD

Next, examine the W' Letters at the positions of the misspellings, Fig. 2-7.

	2	1		2	1																							
	4	3	2	1	0	9	8	7	6	5	4	3	2	1	4	3	2	1	0	9	8	7	6	5	4	3	2	1
W'-1	IARPE D N I A E I E BMSTEACAE														HT E YULWPIESLL													
W'-2	YSHNAM A HTN R ERL B DSLLO														TFEIMEGY E SS S TTTEEF F OHW													

Fig 2-7

Again, read the misspelled letters across by rows in such a way (note that row 2 is first):
W'-2 <- right to left
W'-1 -> left to right

to get the following, PT2-WA, as shown in Fig. 2-8.

F SYTS L AN DAESHES

Fig. 2-8, PT2-WA

Now read the W' letters from Fig. 2-7, going up-down, left to right to get the following, PT2-WD, as shown in Fig 2-9.

NDAAES LST HYES FS

Fig. 2-9, PT2-WD

Part 2.1 - Another One

In part one, an algorithm was given for OTP decryption. Here we will give the corresponding algorithm for OTP encryption.

Assign to each letter of the Kryptos alphabet a number as shown in Fig. 1-6. Then, for a letter of the plaintext and a letter of the pad, add the numbers corresponding to the letters. If the resulting number greater than 25, subtract 26 from it. Take the resulting number and find its corresponding letter. That letter is the letter of the ciphertext.

Examples:

```
plaintext => BE....
pad       => EL....
-----
ciphertext => NY....
```

Example 1:

step 1:

B => 8

E => 11

step 2:

8 + 11 = 19

19 => N

ciphertext => N

Example 2:

step 1:

E => 11

L => 17

step 2:

11 + 17 = 28

step 3:

28 - 26 = 2

2 => Y

ciphertext => Y

Take Row 2 of K4' and Row 6, Columns 1 to 21 of W2' and apply the OTP encryption algorithm for the resulting ciphertext:

K4' R2	FLRVQQPRNGKSSOTWTQSJQ
W2' R6	YETDVTRAHNELNHEESLOPI

	HYOSJXTBASEWZNIBDEENC

The text is in reverse with misspellings (note starting at column 7 the letters: T BASE). Guess the misspellings as follows in Fig 2-10.

P3M	I	S	T
P3	CNEEDBINZWESABTXJSOYH		
W'	IPOLSEEHNLENHARTVDTEY		

Fig. 2-10

which gives us, after adding the ? (the ? starting the letters of K4):

INEEDBINSWESTBTXJSOYH?

Fig. 2-11 PT3

Reading off the letters across left to right, P3M and then W' we get

IST INH

Fig. 2-12 PT3-MW

Interpretations:

What follows are my best guesses - certainly not everything will be correct. CDQ! SOS!

Plaintexts and others:

PT1:

DCI WILL VS BH AWK WYPAN B HZ ERO A-SUNK X
FAN B CQD FIND BY ANORM BERLIN'S D EXPECT
JIM S TO ED GR ROCK IS RD X
ODD RIDDLE 10 I C Q J

PT2:

BR ACT OF LAST LOG A RD Q LRQ
S CNTR L BURIED EYS
SOS JIM S AGAINST LIE Q I RC U X
END Q H PIECES X O ONE P

PT3:

I NEED BINS WEST B TX JS O Y H?

PT1-MA:

NSA BICK Q IO SETS C ED R RD

PT1-MD:

K CDQ IRI BRO DASE SEC NTS

PT1-WA:

EEE ITYM LYE TI DOYTH P HE

PT1-WD:

MY LET HYIPE HET T EIYED O

PT2-MA:

O A LAST D'S MT ECH-E-O

PT2-MD:

DOETH SCE T ALAMO S

PT2-WA:

F SYT'S L AN DAESHES

PT2-WD:

NDAAES LST HYES -FS

PT3-MW:
IS TIN H

Total number of characters: 341, 5 more than K3 and equalling K3 + ?OBKR.

Note I'm guessing the following relations:

K is for C
C is for CIA

D is for T
T is for Teufelsberg, "Devil's Mountain", Berlin
so D is for Devil

PT1:

DCI Will¹ vs BH² awkward weapon B Hz arrow³ a-sunk X
Fan of B CQD⁴ find by abnormal Berlin's D⁵ expect Jim S. to Ed⁶ ground rock is RD⁷ X
Odd riddle - 10 I see⁸? J⁹

1. DCI (Director Central Intelligence) William Webster
2. Berlin Hz Arrow: The big antenna tower that was used by the British GCHQ at the NSA Field Station Teufelsberg, Berlin; see Chaos Computer Club's page: <http://dasalte.ccc.de/teufelsberg/>
3. See the main picture at the site linked to in note 2.
4. CQD is an older radio distress call that was used before SOS
5. D is for Devil - Teufelsberg
6. Jim Sanborn, the sculptor and artist, and CIA cryptographer Edward M. Scheidt
7. Riddle or "Our Devil" meaning our secret listening field site.
6. There are ten pieces to the riddle
9. James Sanborn

PT2:

Brother¹⁰ act of last log a riddle? Location request:
S center L¹¹ buried EYS¹²
SOS Jim S. against lie? I receive you X
End? H pieces X 01 piece.¹³

10. Brother: the British, or perhaps "B R act" meaning "Berlin, our act"

11. S center L: It could mean Kryptos's main sculpture which is shaped like an S, with the L standing for the Log, as in log book - which is a bench. Or the real Log which is part of the sculpture. Or L might stand for lake, as in the pool of water next to the main sculpture, or even the lake between the two rocks with morse code on them. Or should it read S cntrl, short for S central? Even S could just mean "secret".

12. Possibly short for Keys - perhaps the keys are EEEs as in 0xE = 1110 in binary.

13. Beware, I sense deception here!

The others:

NSA BICK Q IO SETS C ED R RD
K CDQ IRI BRO DASE SEC NTS

NSA back? 10 sets, see Ed our riddle
K¹⁴ CDQ our brother's D base¹⁵ secret NTS¹⁶

14. K meaning the CIA

15. D base - Devil's base or Teufelsberg

16. Secret radio intercepts?

EEE ITYM LYE TI DOYTH P HE
MY LET HYIPE HET T EIYED O

O A LAST D'S MT ECH-E-O
DOETH SCE T ALAMO S

F SYT'S L AN DAESHES
NDAAES LST HYES

-FS

EEE item lie T, I doeth protest he
My letter hype, (H)dit (T)dash¹⁷ eyed-o

Oh, a last devil's mountain ech-e-o
Doeth see¹⁸ T Alamo's

Field site's log and dashes
Nada lost his¹⁹

-FS²⁰

17. H is four dits and T is a dash in Morse code

18. Could also mean "secret", or "secret CT" - ciphertext, or "secret see T".

19. Note AES in DAESHES and in NDAAES and that HYES - could mean keys. I think we need to look for some AES Keys! And has anyone noticed how K3 kinda looks like it has been through a simplified version of AES encryption, with an identity subByte, shiftRow, and simplified MixColumns steps.

20. From the Field Site - Or lost his F's, maybe the F's are the keys with 0xF = 1111 in binary.

And:

I NEED BINS WEST B TX JS O Y H? IS TIN H

I need bin's²⁰ west berlin text, Jim S. Oh, why H? Is ten H²¹.

20. Binary or it might refer to JFK's "Ich bin Ein Berliner" speech.

And 21: Why H?

H is the initial of Jim's first name, Herbert,
H is like EEEE in morse code (H = EEEE, S = EEE, I = EE and, of course, E = E),
H is the letter that most looks like 11

Or maybe H is for hexadecimal so

10 is 16! Ziiiiiiiiinnng!

16 parts!? More?

Salam and Happy Hunting,
Ibn Şifre

SUMHGGJKKIPETCTWGTGGPIBZJEJNJKKQRXWCCYXFSBWMJAZUMC